

CROSSVILLE 7 NW, TN, USA

WMO: 754330

Lat: 36.0138N

Lon: 85.1345W

Elev: 583

StdP: 94.51

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 63855

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.3	-9.7	-17.0	0.9	-11.0	-14.4	1.2	-8.3	5.9	6.4	5.6	5.7	1.5	N/A	0.368

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.8	30.4	22.3	29.3	22.3	28.3	22.1	24.0	27.6	23.5	27.1	23.0	26.5	2.0	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.0	19.0	25.8	22.4	18.4	25.3	21.8	17.7	24.8	75.5	27.7	73.3	27.1	71.2	26.7	26.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
5.4	4.7	4.0	-16.3	32.1	3.5	2.4	-18.8	33.8	-20.8	35.2	-22.8	36.6	-25.3	38.3		
			-17.0	25.1	4.0	0.7	-19.9	25.6	-22.3	26.1	-24.5	26.5	-27.5	27.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.7	1.6	3.6	7.9	13.0	17.1	21.1	22.5	22.3	19.2	13.0	7.3	4.0
	DBStd	8.71	6.66	6.13	5.67	4.74	3.89	2.28	2.08	2.17	3.21	4.78	4.94	5.69
	HDD10.0	933	269	192	109	26	3	0	0	0	26	108	200	
	HDD18.3	2478	519	413	325	167	68	5	2	1	29	173	332	444
	CDD10.0	1936	8	12	43	114	223	332	388	382	275	118	26	14
	CDD18.3	439	0	0	1	5	29	87	132	124	54	8	0	0
	CDH23.3	3191	0	0	3	41	213	635	919	906	403	69	2	0
	CDH26.7	700	0	0	0	1	15	127	218	248	82	11	0	0
Wind	WSAvg	1.7	2.2	2.2	2.3	2.1	1.5	1.2	1.2	1.1	1.1	1.4	1.7	2.1
Precipitation	PrecAvg	1520	137	127	158	128	141	122	134	102	105	88	126	151
	PrecMax	1927	279	273	420	286	381	332	337	211	212	220	234	425
	PrecMin	1084	34	27	34	37	47	24	27	38	10	0	45	21
	PrecStd	216	59	56	81	59	69	69	59	41	52	51	47	81
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.9	19.8	23.7	26.2	28.3	30.6	31.9	33.2	30.6	28.0	21.9	19.0
		MCWB	14.8	15.7	15.8	16.9	20.7	20.5	21.9	25.6	21.4	20.1	15.6	16.4
	2%	DB	15.3	17.5	21.4	24.6	26.9	29.1	30.0	30.7	28.8	25.1	19.3	16.9
		MCWB	12.5	15.2	14.5	15.9	19.7	22.1	23.1	22.5	21.0	18.0	13.3	14.3
	5%	DB	13.9	15.5	19.2	23.1	25.7	28.1	28.9	29.2	27.5	23.4	17.4	15.1
		MCWB	11.7	12.5	13.5	15.6	19.1	22.0	22.9	22.5	20.8	17.2	13.2	13.3
	10%	DB	12.0	13.3	17.0	21.5	24.5	27.0	27.9	27.9	25.8	21.3	15.7	13.0
		MCWB	10.1	10.4	12.4	14.9	18.5	21.5	22.6	22.1	20.4	16.6	12.6	11.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.6	17.0	16.8	19.5	21.9	24.1	24.9	24.9	23.0	21.8	18.0	17.5
		MCDB	16.7	18.8	21.1	23.2	26.5	28.0	28.2	28.9	26.9	25.1	19.6	18.6
	2%	WB	13.7	15.9	15.8	18.1	20.9	23.4	24.1	23.9	22.4	20.6	16.4	15.4
		MCDB	14.5	17.6	19.0	22.4	25.0	27.5	27.6	27.3	26.2	23.5	17.8	16.4
	5%	WB	12.0	14.3	14.8	17.1	20.3	22.8	23.7	23.3	21.8	19.0	15.1	13.5
		MCDB	13.1	16.0	17.9	21.1	24.2	26.7	27.2	26.7	25.5	21.6	16.1	14.8
	10%	WB	9.8	11.9	13.4	16.2	19.6	22.1	23.1	22.7	21.0	17.4	12.7	11.5
		MCDB	10.8	13.2	16.2	19.9	23.4	25.4	26.6	25.9	24.4	19.9	14.6	12.9
Mean Daily Temperature Range	5% DB	MDBR	9.3	9.7	10.6	11.6	11.1	10.7	9.8	10.2	11.0	11.9	11.4	9.2
		MCDBR	9.7	11.2	12.9	13.3	12.6	12.1	11.1	12.2	13.2	13.8	13.0	9.9
		MCWBR	7.8	7.5	7.8	6.9	5.9	5.6	4.7	5.3	5.8	7.1	7.6	7.2
	5% WB	MCDBR	8.4	8.8	10.4	10.5	10.4	10.0	9.4	9.5	10.5	10.9	7.8	8.3
		MCWBR	7.6	7.2	7.2	6.2	5.5	5.5	4.7	4.9	5.2	6.8	6.5	7.3
Clear-Sky Solar Irradiance	taub	0.299	0.311	0.338	0.374	0.419	0.449	0.478	0.486	0.393	0.353	0.316	0.296	
	taud	2.525	2.482	2.441	2.329	2.229	2.172	2.120	2.083	2.328	2.441	2.508	2.549	
	Ebn at Noon	906	932	931	909	868	838	811	795	861	871	875	887	
	Edh at Noon	80	93	106	125	141	149	156	159	118	96	80	73	
All-Sky Solar Radiation	RadAvg	2.21	2.77	3.92	5.12	5.67	6.14	5.81	5.44	4.60	3.64	2.62	1.91	
	RadStd	0.20	0.35	0.38	0.36	0.46	0.40	0.33	0.28	0.44	0.47	0.29	0.18	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (24 neighbors)	+0.46	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+137	+92

Nomenclature: See separate page